

Global Renewable Energy Adoption: A 2026 Research Brief

1. Executive Summary

This brief summarizes recent trends in renewable energy adoption across major economies. It draws on public utility filings, national energy agency reports, and industry association data to characterize growth in solar, wind, and battery storage capacity between 2020 and 2026.

2. Global Capacity Trends

Renewable sources accounted for an estimated 34 percent of global electricity generation in 2026, up from 27 percent in 2020. Solar photovoltaic capacity grew the fastest among all generation types, with average annual installed capacity increasing by roughly 22 percent per year over the period. Onshore wind capacity grew more modestly, at an average of 9 percent per year, while offshore wind expanded from a small base at an average annual rate of 18 percent.

2.1 Regional Variation

Growth was uneven across regions. East Asia led in absolute installed capacity additions, driven largely by utility-scale solar. Northern Europe led in offshore wind deployment, supported by shallow continental shelf waters and established grid interconnection. North America saw the fastest growth in battery storage paired with solar, driven in part by tax incentives introduced in the early 2020s.

3. Battery Storage

Grid-scale battery storage capacity roughly quadrupled between 2020 and 2026.

Lithium-iron-phosphate chemistries became the dominant technology for stationary storage due to lower cost and improved safety characteristics relative to earlier nickel-based chemistries. Average storage duration for newly installed systems increased from about 2 hours to just over 4 hours over the same period.

4. Policy Drivers

Three policy mechanisms were most frequently cited across the reports reviewed: production tax credits, renewable portfolio standards requiring utilities to source a minimum share of electricity from renewables, and streamlined permitting for transmission infrastructure. Analysts noted that permitting delays for new transmission lines remained the most commonly cited bottleneck limiting faster renewable deployment.

5. Outlook

Most forecasts reviewed for this brief project renewables reaching between 40 and 45 percent of global electricity generation by 2030, contingent on continued cost declines in storage and sustained policy

support. The primary downside risks identified were supply chain constraints for critical minerals and slower-than-expected grid modernization.